

SHAHRIAR RAHMAN

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Objectives

AI/ML-focused software engineer with a published deep learning paper (Springer 2026) and hands-on experience building ML-integrated systems in production. Skilled in computer vision, human computer interaction, anomaly detection, and full-stack AI application development. Seeking a fully funded thesis-based MSc in Canada to advance research in Applied AI or Machine Learning, with a long-term goal of joining a top-tier AI/ML engineering team.

Education

Bachelor of Science in Computer Science and Engineering

2021 – 2025

United International University (UIU) — Dhaka, Bangladesh

CGPA: 3.46 / 4.00 — Equivalent to 83–84% on a percentage of UIU scale

Publications

Akter, T., **Rahman, S.**, Billah, A., Haque, A., & Rahman, R. (2026). *A Deep Learning Approach for Indoor Plant Health Monitoring: Classification of Healthy, Unhealthy, and Dead Plants*. In: ICT for Intelligent Systems — ICTIS 2025. Lecture Notes in Networks and Systems, vol. 1510. Springer, Singapore.

https://doi.org/10.1007/978-981-96-9275-0_14

Work Experience

Full-Stack Developer

2025 – Present

AvranceCorp Developments — Toronto, Canada

- Built TradeREA, a real estate tokenization platform, pretty much from scratch — owning the full stack across an investor portal, admin panel, live trading interface, and KYC/compliance workflows, all the way from design to deployment.
- Put a lot of thought into the UI/UX in Next.js 14 and Tailwind CSS — translating genuinely complex financial data like real-time order books, portfolio analytics, and dividend tracking into something non-technical investors could navigate without getting lost.
- Built an ML anomaly detection system for real-time suspicious login identification using Isolation Forest and supervised classifiers — a tricky problem given how little labeled fraud data you typically have in financial platforms.
- Integrated NorthCapital's compliance API (Reg D 506(c)) for identity verification and securities offering management — involved navigating regulatory constraints alongside some genuinely interesting systems design challenges.
- Set up multi-role authentication with JWT session revocation, 2FA, and real-time threat scoring — tried to keep it genuinely secure without making the experience feel hostile for normal users.

Project Work

ML-Based Login & Auth Fraud Detection

Jan – Feb 2026

Python · Isolation Forest · Scikit-learn · Node.js · Express.js

- Built and deployed an anomaly detection system to flag suspicious login attempts on a live crypto exchange — Isolation Forest doing the heavy lifting, integrated directly into the production backend with automated alerting.

Network Intrusion Detection System

Jun – Oct 2025

[Python](#) · [CNN](#) · [Random Forest](#) · [SVM](#) · [K-NN](#) · [Scikit-learn](#)

- Put together an ensemble ML system — CNN, Random Forest, SVM, and K-NN — to catch network intrusions on real-time traffic. The interesting part was getting the false positive rate down without sacrificing speed.

Job Scraping & Smart Categorisation Pipeline

Jan – Nov 2025

[Node.js](#) · [Puppeteer](#) · [Express.js](#) · [React.js](#) · [LLMs](#)

- Scraped job listings from multiple sources and used an LLM to automatically tag each one by role, company, and location — saved a lot of manual sorting and was a genuinely fun applied NLP problem to solve.

Indoor Plant Health Detection — Published Research

Oct – Dec 2024

[Python](#) · [YOLOv8–v11](#) · [PyTorch](#) · [TensorFlow](#)

- Led the deep learning side of a research project classifying plant health from around 1,300 images. Ran a comparison across YOLOv8 through YOLOv11 with majority-vote ensemble — YOLOv11 came out best, and the work ended up published in Springer.

Student Resources & Services Platform

Jan – Apr 2024

[React.js](#) · [Node.js](#) · [Express.js](#) · [MySQL](#)

- A full-stack web platform built with a small team for fellow students — exam resource search, a digital library, and a housing finder all in one place.

Technical Skills

AI / ML: PyTorch, TensorFlow, Scikit-learn, HuggingFace, YOLO, Computer Vision, Anomaly Detection, NLP, LLMs

Languages: Python, JavaScript, C/C++, Java

Web & Backend: React.js, Next.js, Node.js, Express.js, PostgreSQL, MySQL, REST APIs

Cloud & Tools: AWS (S3, EC2), Git, Docker, Figma, TailwindCSS

Extracurricular Involvement

Member — UIU Computer Club (UIUCCL)

2022 – 2024

United International University — Dhaka

- Took part in workshops on Android dev, UX/UI design, and web development — including Google's Android Study Jam and a Design Thinking Workshop organised by the club.

Member — UIU Robotics Club

2022 – 2023

United International University — Dhaka

- Finished the Electronics & Basic Robotics track, collaborated on a few hardware builds, and joined the 'Outside the Circuits' STEM webinar series.

Volunteer — CSE Project Show

2022 – 2024

United International University — Dhaka

- Helped run the annual student project exhibition across multiple years — coordinating with teams, setting up displays, and guiding industry guests through the showcase.